

[Art of Problem Solving](#)[AoPS Online](#)[Beast Academy](#)[AoPS Academy](#)

Euclidean Geometry In Mathematical Olympiads

[Bookmark](#) [New Topic](#)

Problem 1.36



Chapter 1

fgm



Bookmark

Reply

Malstern

198 posts

Dec 31, 2018, 12:04 PM • 2 👍

#1

Let $ABCDE$ be a convex pentagon such that $BCDE$ is a square with center O and $\angle A = 90^\circ$. Prove that AO bisects $\angle BAE$.

NikolsLife

9657 posts

Jan 1, 2019, 7:38 AM • 1 👍

#2

Hint



Try to find a cyclic quadrilateral.

Solution



Since $\angle BAE = 90^\circ$ and $\angle BOE = 90^\circ$, this means A, B, O, E are concyclic.

$\angle BAO = \angle BEO = 45^\circ$ and $\angle EAO = \angle EBO = 45^\circ$, so $\angle BAO = \angle EAO$, which proves that AO bisects $\angle BAE$.

This post has been edited 1 time. Last edited by NikolsLife, Jan 1, 2019, 7:39 AM

USER394745

116 posts

Jul 17, 2019, 4:31 PM • 2 👍

#3

Click to reveal hidden text



Or you could say that since O is the center, $BO=OE$ and therefore $\text{arc}(BO)=\text{arc}(EO)$ and thus the two angles are equal.

bever209

1522 posts

Apr 12, 2020, 6:43 PM

#4

I just want to say that there is a very similar question (basically the same actually) in Volume 2 chapter 3

Archimedes15

1490 posts

Jun 24, 2020, 6:27 PM • 1 👍

#5

alright before I forget to copy-paste off word doc, here goes:

Solution

Jerry122805

1074 posts

Aug 12, 2020, 4:46 PM

#7

IK this is redundant but just to practice my latex

Solution



We connect B to O and E to O to obtain $\angle BOE = 90^\circ$. Therefore we see $ABOE$ is a cyclic quadrilateral since $\angle BOE = \angle BAE = 90^\circ$. Now we see $\angle EOB = 45^\circ$ since O is the center of $BCDE$. Therefore by cyclic $ABOE$ we have $\angle OEB = \angle OAB = 45^\circ \implies \angle OAE = 45^\circ$ and thus OA is an angle bisector.

Also can someone check my solution with directed angles; I'm relatively new to using them

and I wanna give them a try.

Solution with directed angles



We know that $\angle BOE = \angle BAE = 90^\circ$ and so $BOEA$ is a cyclic quad. We also know that $\angle OEB = \angle OAB = 45^\circ$ so OA indeed bisects $\angle BAE$.

This post has been edited 4 times. Last edited by Jerry122805, Aug 13, 2020, 5:00 PM

new_to_mew_t...
65 posts

May 24, 2021, 12:28 AM

#8

Radical Axis Solution

Hypatia1728
728 posts

Feb 18, 2022, 7:13 PM

#9

similar sol, just for practice

franzliszt

Jul 15, 2022, 1:21 PM • 2 👍

#10

Reflect A over O and construct a square: